INCH-POUND

MIL-A-0083216C(ME)
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USED IN LIEU OF
MIL-A-83216B(USAF)
25 May 1976

MILITARY SPECIFICATION

AIR CONDITIONER, HORIZONTAL,

54,000 BTUH, 208/230V

This specification is approved for use by the US Army Belvoir Research, Development, and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-A-83216B(USAF). It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers the requirements for a 208/230V, 3 phase, 50/60 Hz, electric-motor-driven military air conditioner, having a nominal cooling capacity of 54,000 British thermal units per hour (Btu/hr), heating capacity of 31,000 Btu/hr, and providing dehumidification and fresh air.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications and standards form a part of this document to the intent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4120

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SPECIFICATIONS

FEDERAL

BB-F-1421	- Fluorocarbon Refrigerants.
QQ-B-654	- Brazing Alloys, Silver.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronics and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-100	- Engineering Drawing Practices.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-882	- System Safety Program Requirements.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Military Materiel (Metric).

HANDBOOKS

MILITARY

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

ARMY

TA13230E3500 - Air Conditioner: Horizontal, 54,000 BTUH, 208/230 Volts, Three Phase, 50/60 Hertz.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: USA BELVOIR RDE CTR, ATTN SATBE FE, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AIR MOVING AND AIR CONDITIONING ASSOCIATION, INC. (AMCA)

Standard 210 - Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to the AIR MOVE & COND ASSN INC, 30 W UNIV DR, ARLINGTON HEIGHTS IL 60004.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS, INC. (ASHRAE)

STD 37 - Methods of Testing for Rating Unitary Air Conditioning and Heat Pump Equipment.

STD 51 - Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to the AMERCN SCTY OF HTG, REFRIG, & AIR COND ENGR, 1791 TULLIE CIR, NE, ATLANTA GA 30329.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 280 - Specification for Seamless Copper Tube for Air Conditioning and Refrigeration Field Service.

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F 872 - Filter Units, Air Conditioning: Viscous-Impingement Type, Cleanable.
D 3951 - Commercial Packaging.

(Application for copies should be addressed to the AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT (SCAQMCD)

Rule 442 - Usage of Solvents

(Application for copies should be addressed to: S MGMT CNTRL DIST, 9150 E FLAIR DR, EL MONTE CA 91731.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The air conditioners shall be in accordance with TA13230E3500 and as specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the air conditioners (see 6.2). Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or the designated representative.

3.4 Reliability and maintainability.

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3.4.1 Reliability. When tested as specified in 4.5.2.17, the mean-time-between-failure shall be as follows:

Upper test MTBF (θ_0) = 960 hours

Lower test MTBF (θ_1) = 480 hours

3.4.2 Maintainability.

3.4.2.1 Maintenance ratio. When tested as specified in 4.5.2.18, the air conditioners shall have a maintenance ratio of not more than 0.03. The maintenance ratio is defined as the ratio of the total active maintenance manhours required (scheduled and unscheduled) to the total operating time. Manhours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing. Not more than 25 percent of repairs shall require maintenance at the general support level.

3.4.2.2 Scheduled maintenance. When tested as specified in 4.5.2.18, the air conditioners shall not require scheduled maintenance more often than every 250 hours of operation. Scheduled maintenance shall not be required at any level higher than organizational maintenance.

3.5 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor, and shall be subject to all provisions of this specification.

3.5.1 Material deterioration prevention and control. The air conditioners shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the air conditioners may be exposed.

3.5.1.1 Fungusproof materials. Materials that are nutrients of fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.5.1.2 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service. Cadmium plating/plated items shall not be used.

3.5.1.2.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals and methods of protection are defined MIL-STD-889.

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3.5.1.3 Gasketing and insulating materials. Rubber or similar gasketing and insulating materials shall be compounded to ensure oil resistance and, where applicable, ability to withstand low temperatures.

3.5.1.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the air conditioners may be newly fabricated from recovered materials to the maximum extent practicable, provided the air conditioners produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the air conditioners.

3.6 Performance.

3.6.1 Ambient operating range. The air conditioner shall be capable of starting and supplying refrigerated, recirculated air within the range of ambient conditions specified in 3.6.7.2. The cooling capacity of the air conditioner shall be in accordance with 3.6.2.

3.6.1.1 Extreme operating conditions. The air conditioner shall have the capability to start and operate continuously in the cooling mode with air entering and being maintained to both the evaporator and condenser at 51.7 °C (125 °F). In addition, the air conditioner shall have the capability to start and operate continuously in the cooling mode under the conditions specified in 3.6.2.1.

3.6.2 Cooling capacity. The minimum cooling capacities at sea level of the air conditioners shall be as indicated in table I. The air conditioner shall be operated against 0-inch water gauge (in. wg) static pressure during capacity testing.

TABLE I. Capacity rating conditions.

	Ambient Temp °C (°F)	Evaporator Inlet Temp °C (°F)		Minimum Net Total Cooling Capacity
Condition	Dry Bulb	Dry Bulb	Wet Bulb	(Btu/hr)
1	46.1 (115)	29.4 (85)	21.1 (70)	54,000
2	51.7 (125)	32.2 (90)	18.3 (65)	50,000
3	35.0 (95)	26.7 (80)	19.4 (67)	60,000
4	10.0 (50)	26.7 (80)	14.4 (58)	70,000

3.6.2.1 Cooling capacity at 10 °C (50 °F) ambient. The cooling capacity of the air conditioner shall be demonstrated at condition 4 of table I.

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3.6.3 Evaporator air flow. When the air conditioner is operating under condition 1 of table I, the evaporator fan shall provide a minimum of 2,200 cubic feet per minute (cfm) at 0-inch wg and a minimum of 2,000 cfm against a static pressure of 1.0-in. wg.

3.6.3.1 Fresh air. The air conditioner shall be capable of supplying at least 200 cfm of filtered, cooled, fresh (outside) air when operating against an external static pressure of 1.0-in. wg. The 200 cfm may be part of the air flow requirements of 3.6.3 when operating in the vent mode. The amount of fresh air shall be controlled by a manual damper.

3.6.4 Heating capacity. The air conditioner shall provide a minimum heating capacity of 16,800 Btu/hr in the low heat mode and 31,000 Btu/hr in the high heat mode in an ambient temperature of 65 to 80 °F.

3.6.5 Dehumidification. The air conditioner shall demonstrate the capability of removing moisture from the conditioned air when operated in the conditions of table II.

TABLE II. Dehumidification conditions.

Condition	Ambient Air Temperature (Dry-Bulb) °C (°F)	Return Air Temperature	
		°C (°F) (Dry-Bulb)	°C (°F) (Wet-Bulb)
1	46.1 (115)	32.2 (90)	29.4 (85)
2	46.1 (115)	26.7 (80)	23.3 (74)
3	46.1 (115)	23.9 (75)	20.6 (69)
4	46.1 (115)	21.1 (70)	18.3 (65)

3.6.6 Power consumption. When the air conditioner is operated under condition 1 of table I or in high heat mode of 3.6.4, the total power consumption shall not exceed 10.5 kw.

3.6.7 Environmental. The air conditioners shall be capable of withstanding exposure to the following conditions:

3.6.7.1 Storage conditions. The air conditioners shall withstand storage temperatures from -60 to +160 °F without impairment of specified capabilities.

3.6.7.2 Operating conditions. The air conditioners shall start and operate as specified under the following conditions:

a. Temperatures from 50 to 125 °F in the cooling mode.
b. Temperatures from 50 to 85 °F in the heating mode.
c. Rainfall as specified in 3.6.7.4.

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3.6.7.3 Humidity and salt fog. When subjected to the tests specified in 4.5.2.13 and 4.5.2.14, no part of the air conditioner shall show evidence of deterioration, corrosion, or change in tolerance limits that could prevent the unit from meeting operational requirements. The unit shall show no evidence of peeling of paint, malfunctioning, refrigerant leaks, rough operation, electrical shorts, erratic pressure, or other irregular operation.

3.6.7.4 Rain. When subjected to the tests specified in 4.5.2.12, the air conditioners shall operate without functional damage or impairment of operation. There shall be no penetration of water from the evaporator section to the dry side of the enclosure or into or on any electrical component; in particular, the motors, electrical connectors, terminals switches, or controls that could result in an electrical short, cause damage from rust or corrosion, or become a hazard to personnel.

3.6.7.5 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be treated for the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.6.7.5.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish. Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces. Surfaces shall be cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.6.7.5.1 Varnish composition. Composition I shall be used unless composition II must be used to comply with local air-pollution regulations in the application of varnish. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with South Coast Air Quality Management Control District Rule 442 - Usage of Solvents. The certification shall be made available for review by the contracting officer.

3.6.7.6 Vibration. When subjected to the test specified in 4.5.2.15, the air conditioners shall show no evidence of structural damage, misalignment or malfunction of components, leaks, fractures, abnormal vibration, or any other irregular operation.

3.7 Air system.

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3.7.1 Fan and blower assemblies. Fan and blower assemblies shall deliver the required quantities of air under all specified conditions. The fans and blowers shall be constructed and rated in accordance with AMCA Bulletin 210. Fans shall be statically and dynamically balanced.

3.7.2 Air filters. Filtration of the evaporator air shall be accomplished through the use of filters conforming to ASTM F 872 except, the initial dust-arresting efficiency shall be not less than 60 percent at 500 feet per minute (fpm) net face velocity.

3.7.3 Duct adapters. The air conditioner shall be furnished with insulated duct adapters and protective end covers. The air conditioner shall permit storage of the protective end covers in the condenser section without causing damage or performance degradation when the air conditioner is set up for normal operation.

3.7.4 Flexible ducts. Flexible ducts shall be furnished with each air conditioner.

3.8 Refrigeration system.

3.8.1 Refrigeration system type. The refrigeration system shall be of vapor-cycle type employing refrigerant conforming to BB-F-1421, type 22, and shall be furnished with a full charge in the system.

3.8.2 Moisture content. The moisture content in the system shall be not more than 25 parts per million (ppm) by weight in the liquid phase.

3.8.3 Refrigerant system pressure and leakage. When tested as specified in 4.5.2.3.2, the entire refrigerant system shall withstand pressures up to 425 psig without deformation of any components or any detectable leaks.

3.8.4 Refrigerant piping. All copper refrigerant piping shall conform to ASTM B 280. The tubing shall be fabricated with the use of high-temperature solder conforming to QQ-B-654.

3.8.5 Compressor. While operating at an ambient temperature of 51.7 °C (125 °F) and with return air at 51.7 °C (125 °F), the compressor shall be a hermetically sealed type and have adequate cooling capacity to meet the requirements specified herein without overloading the compressor motor. The compressor shall be provided with a crankcase heater.

3.8.6 Condenser. The condenser shall be of an air-cooled copper tube-aluminum fin type and shall withstand, without deformation or leakage, internal pressures as specified.

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3.8.7 Evaporator. The evaporator shall be of the copper tube-aluminum fin type and withstand, without deformation or leakage, internal pressures as specified.

3.8.8 Charging valves. High and low side charging valves shall be provided conforming to Drawing 13216E6168.

3.8.9 Moisture indicator. A moisture indicator shall be provided to detect moisture in the refrigerant while the air conditioner is in operation or the system is being charged.

3.9 Control systems.

3.9.1 Safety. The following safety control features shall be provided on the air conditioner:

a. High pressure cutout switch and pressure relief valve in refrigerant system.
b. Thermal protection for compressor and fan motors.
c. Reset relay to prevent unit cycling on safety controls.
d. Fusible plug.
e. High temperature cutout for electric heaters.
f. Circuit breaker for control circuit.

3.9.2 Operator controls. Operator controls as specified below shall be installed on a single control panel except the manual damper control for fresh air.

3.9.2.1 Control switch. Off/reset, vent and automatic cooling/heating modes shall be provided. The evaporator fan shall operate continuously in the vent and automatic cooling/heating modes. With the switch in the "OFF" position and with power connected to the air conditioner, the compressor crankcase heater shall be energized.

3.9.2.2 Conditioned air thermostat. A cooling/heating thermostat shall be provided. There shall be a 8.3 °C (15 °F) temperature differential between cooling off and heating on cycles. Two-step heating control shall be provided with a temperature differential of 1.6 ±0.55 °C (3 ±1 °F) between low and high heat stages. A mercury switch-type thermostat shall not be provided.

3.9.2.3 Circuit breaker. A circuit breaker, with reset for the control circuit, shall be provided.

3.10 Electrical system.

3.10.1 Electric power supply. The air conditioner shall operate as specified from a 208/230V, 3 phase, 50/60 Hertz, 4 wire source.

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3.10.2 Safety. The air conditioner shall be in conformance with MIL-STD-454, requirement 1.

3.10.3 Number coding. All wiring shall be number-coded. Wire 1/4-inch and under in diameter shall be coded on both sides diametrically opposite at intervals varying from 1 to 6 inches center-to-center. Wire over 1/4 inch in overall diameter shall be marked by cable markers at each detachable end.

3.10.4 Terminations. Before being soldered to terminals, wires shall be effectively secured to the terminals so that the connections are not dependent on soldering alone for mechanical strength. Varnish or lacquer conforming to MIL-V-173 as specified for class II materials shall be applied to the end of fibrous coverings, binders, and similar items that are not otherwise secured against fraying.

3.10.5 Terminals. Terminals attached to wires may be of either the solder or solderless type, except for the following limitations on the solderless type:

3.10.6 Solderless terminals. Solderless terminals shall have a means for gripping the insulation. The terminals shall be used on stranded conductors ranking from 642.6 to 10380 circular mils (No. 22 to No. 10 American wire gauge [AWG]).

3.10.7 Relays. The relays shall be capable of operating under the service conditions specified.

3.10.7.1 Relay coils. Except for relays that are hermetically sealed, the coils shall be vacuum impregnated with varnish conforming to MIL-V-173.

3.10.7.2 Time delay relay. A nominal 1-minute time delay shall be provided which will delay compressor operation only.

3.10.7.3 Reset relay. A reset relay shall be provided which will prevent unit cycling when any safety control is activated.

3.10.8 Power cable. The air conditioner shall be provided with a power cable conforming to TA13230E3500.

3.11 Housing.

3.11.1 Enclosure. The complete air conditioner shall be fully enclosed in a metallic enclosure. The enclosure shall be provided with drain channels or constructed to prevent the entry of rain or melting snow. The evaporator section shall be internally insulated to prevent condensation.

3.11.2 Condensation drain and pan. A condensation drain and pan shall be provided in the evaporator section to collect and dispose of moisture condensed by the evaporator coils.

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3.11.3 Condensate drain hoses. Condensate drain hoses and all necessary fittings shall be provided to carry away condensed moisture. The hoses and fittings shall not be damaged by exposure to the temperatures as specified in 3.6.7.

3.11.4 Stacking bolts. The air conditioner shall be provided with stacking bolts to permit up to three units to be stacked and secured.

3.12 Human factors engineering. The air conditioner design shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), controls (5.4), labeling (5.5), work space design requirements (5.7), design for maintainability (5.9), small systems and equipment (5.11), and hazards and safety (5.13) of MIL-STD-1472 as applicable.

3.12.1 Noise levels. When measured as specified in 4.5.2.21, the noise levels produced by the air conditioner shall not exceed the sound pressure level values specified in table III.

TABLE III. Noise level requirements.

Octave Band Center Frequencies (Hz)	Sound Pressure Level (dB ref. 0.0002 dynes/cm^2)
63	75
125	70
250	66
500	66
1000	65
2000	62
4000	54
8000	49

3.13 Dimensions. The dimensions of the air conditioner shall not exceed those specified in TA13230E3500.

3.14 Weight. The weight of the complete air conditioner including duct adapters, refrigerant, oil, power cable, and flexible ducts shall not exceed 900 pounds.

3.15 Treatment and painting. All parts shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable, with the following exceptions:

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a. Internal components previously painted by the manufacturer need not be repainted when they conform to the salt/fog requirements.
b. Coils, refrigerant tubing, electrical components, or other components whose operation will be impaired by painting shall not be painted.

3.15.1 Finish. Unless otherwise specified (see 6.2), the top coat shall be camouflage green 383 in accordance with MIL-C-46168.

3.16 Safety. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such a nature or are so located as to be a hazard to operating personnel shall be guarded or insulated to the extent necessary to eliminate the hazard. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

3.17 Operating instructions. Operating instructions, refrigeration and wiring diagrams of the air conditioner shall be provided. The instructions and diagrams shall be such that oil, dirt, water, et cetera, will not affect their legibility. The wiring diagram shall conform to MIL-STD-100.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.19 Workmanship. The air conditioner, including all parts and accessories, shall be fabricated and finished in accordance with good commercial practices. Particular attention shall be given to freedom from blemishes, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts; and tightness of assembly screws and bolts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor's own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government, may be used. The Government reserves the right to perform any of the inspections set forth in the specification

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where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. Failure of any test or examination of the first article shall be cause for disassembly of the first article, in the presence of a Government representative, to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and drawings referenced with regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be cause for rejecting the first article.

4.1.3 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examinations. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection and cause for performing the inspection specified in 4.1.2 as applicable.

4.3.2 First article tests. The first article air conditioners shall be subjected to the test marked "X" in columns 2 through 6 of table IV in the sequence listed. Failure of any test shall be cause for rejection and performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

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4.4.1 Examination. Each air conditioner shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each air conditioner shall be subjected to the tests marked "X" in column 7 of table IV in the sequence listed. Failure of any test shall be cause for rejection.

4.5 Examination procedures.

4.5.1 Examination. The air conditioner shall be inspected for the following defects:

101. Parts or components missing or not as specified (see 3.3 and 3.7 thru 3.11.4). Each part or component missing or not as specified shall constitute one defect.
102. Housing not insulated (see 3.11.1).
103. Bolted and riveted connections not as specified (see 3.3 and 3.19).
104. Parts hazardous to operating and maintenance personnel not fully enclosed, insulated, or properly guarded (see 3.10.2, 3.12, 3.16, and 3.19).
105. Material not as specified (see 3.5).
106. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.5.1).
107. Materials not fungusproof or treated to be fungusproof as specified (see 3.5.1.1).
108. Metals not corrosion-resistant or treated to resist corrosion as specified (see 3.5.1.2).
109. Cadmium plating used (see 3.5.1.2).
110. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.5.1.2.1).
111. Gasketing and insulating materials not compounded to ensure their oil resistance and, where applicable, to withstand low temperatures (see 3.5.1.3).
112. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the air conditioner (see 3.5.1.4).
113. Electric power cable not as specified (see 3.10.8).
114. Fans not statically and dynamically balanced (see 3.7.1).
115. Safety control features not as specified (see 3.9.1).
116. Control panel not as specified or control devices missing (see 3.9.2 thru 3.9.2.3).
117. Wiring requirements or wiring marking not as specified (see 3.10.3 and 3.10.4).
118. Terminals not as specified (see 3.10.5 and 3.10.6).
119. Relays not as specified (see 3.10.7 thru 3.10.7.3).
120. Housing not as specified (see 3.11.1 thru 3.11.4).
121. Human factors engineering not as specified (see 3.12).
122. Dimensions exceed those specified (see 3.13).
123. Total weight of the unit fully charged and ready to operate exceeds the weight specified (see 3.14).

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124. Treatment and painting not as specified (see 3.15 and 3.15.1).
125. Flexible ducts not as specified (see 3.7.4).
126. Operating instructions not as specified (see 3.17).
127. Marking and identification not as specified (see 3.18).
128. Welding not as specified (see 3.19).
129. Workmanship not as specified (see 3.19).

4.5.2 Tests.

4.5.2.1 Test schedule.

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TABLE IV. Test schedule. 1/

Test Req	First article test number					Qual conf	Test	Test para	Reqmnt Para
	1	2	3	4	5				
1	2	3	4	5	6	7	8	9	10
1	X	X	X	X	X	X	Refrigeration system pressure.	4.5.2.3.1	3.8.3
2	X	X	X	X	X	X	Examination.	4.5.1	3,1. 3.3, 3.7, 3.9, 3.10, 3.11, 3.12, 3.13, 3.14, 3.15, 3.17, 3.18, 3.19
3	X	X	X	X	X	X	Controls.	4.5.2.4	3.9.2.1, 3.9.2.2, 3.9.2.3
4	X	-	-	-	-	-	Refrigerant moisture content.	4.5.2.5	3.8.2
5	X	-	-	-	-	-	Cooling capacity and airflow.	4.5.2.6	3.6.2, 3.6.3, 3.6.6
6	X	-	-	-	-	-	Heating capacity.	4.5.2.7	3.6.4
7	-	X	-	-	-	-	High temperature operation, cooling.	4.5.2.8	3.6.1.1, 3.6.7.2
8	-	X	-	-	-	-	High temperature storage.	4.5.2.9	3.6.7.1
9	-	X	-	-	-	-	Low temperature operation cooling.	4.5.2.10	3.6.2.1, 3.6.7.2
10	-	X	-	-	-	-	Low temperature storage.	4.5.2.11	3.6.7.1
11	-	X	-	-	-	-	Rain.	4.5.2.12	3.6.7.4
12	X	-	-	-	-	-	Humidity.	4.5.2.13	3.6.7.3

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TABLE IV. Test schedule. 1/ (Continued)

Test Req	First article test number					Qual conf	Test	Test para	Reqmnt Para
	1	2	3	4	5				
1	2	3	4	5	6	7	8	9	10
13	-	X	-	-	-	-	Salt fog.	4.5.2.14	3.6.7.3
14	-	X	-	-	-	-	Vibration.	4.5.2.15	3.6.7.6
15	X	X	-	-	-	X	Run-in.	4.5.2.16	3.2
16	X	X	-	-	-	X	Refrigeration system leak.	4.5.2.3.2	3.8.3
17	-	-	X	X	X	-	Reliability.	4.5.2.17	3.4
18	-	-	X	X	X	-	Maintenance evaluation.	4.5.2.18	3.4.2.1, 3.4.2.2
19	X	X	-	-	-	-	Safety.	4.5.2.19	3.16
20	X	-	-	-	-	-	Human factors engineering.	4.5.2.20	3.12
21	X	-	-	-	-	-	Noise.	4.5.2.21	3.12.1

1/ Perform inspection marked "X".

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4.5.2.2 Conditions. Unless otherwise specified herein, tests shall be conducted using 208V, 60-Hz input electrical supply. Unless otherwise specified herein, no braces, insulating or sealing materials, tiedown devices, or other items that are not inherently a part of the operating air conditioner shall be employed. Unless otherwise specified herein, no alignment of components, addition of refrigerant, or other servicing shall be performed on the preproduction or production air conditioners after testing of the air conditioner has begun. Components subject to atmospheric corrosion shall not be treated with any treatment such as grease or oil to meet test requirements.

4.5.2.3 Refrigerant system pressure and leakage.

4.5.2.3.1 Refrigerant system pressure. Charge the refrigerant system to 425 psi, ±5 psig or -0 psig with dry nitrogen containing refrigerant gas detectable with a halogen-sensitive electronic leak detector (General Electric type H-2 or equal). Reduce the pressure to 300 psig, ±5 psig. Calibrate the leak detector for the type of gas and percentage of mixture employed and set the detector to indicate a leakage rate of 0.1 ounce per year corrected to the percentage of the mixture. Check the entire refrigerant system for leaks. One or more of the following shall constitute failure of this test:

a. Permanent deformation of any component in the refrigerant system.
b. Any detectable refrigerant leak.
c. Nonconformance to 3.8.3.

4.5.2.3.2 Refrigerant system leakage. Test for leakage at room ambient temperature (70 °F or above) with a halogen-sensitive electronic leak detector (General Electric Type H-2 or equal). Set the detector to indicate a leakage rate of 0.1 ounce per year. Check the entire refrigerant system for leaks. A detectable leak shall constitute failure of this test.

4.5.2.4 Controls. Electrical controls shall be tested as specified herein. With power connected to the air conditioner, position the mode selector switch and thermostat in the following position:

a. "OFF" position. Determine that the air conditioner components have been de-energized as specified in 3.9.2.1 and the compressor crankcase heater is energized.
b. "VENTILATE" position. Determine that the evaporator fan motor is energized and operating, and that the compressor motor, condenser fan motor and the heaters are off.
c. Position-thermostat knob at 12 o'clock position and mode selector switch on "AUTO". Immerse the conditioned-air thermostat bulb in water or air having a temperature of 85 +5 -0 °F. Determine that the evaporator and condenser fan motors are energized and operating and the heaters are off. After a time delay, determine that the compressor

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motor has started and is operating. Determine that there is refrigerant flow through the liquid indicator to the expansion valve.

d. Immerse the conditioned-air thermostat bulb in water or air having a temperature not exceeding 70 +1 -0 °F. Determine that the compressor motor and evaporator and condenser fan motors are de-energized. Determine that there is no refrigerant flow through the liquid indicator to the expansion valve.

e. Immerse the conditioned-air thermostat bulb in water or air having a temperature of 67.5 +0 -1 °F. Determine that the evaporator fan motor is energized and operating and the compressor motor and condenser fan motor are off. Determine that both stages of heaters are energized and operating.

Nonconformance to 3.9.2.1 or 3.9.2.2 shall constitute failure of this test.

4.5.2.5 Refrigerant moisture content. After not less than 1 hour of continuous operation, the cooling cycle in an ambient temperature of not less than 95 °F, shut down the air conditioner. After an additional period of 1 hour, withdraw a sample of approximately 1 pound of refrigerant from the low-pressure side and determine the moisture content by one of the methods specified in BB-F-1421 or by the titrimetic method using the Karl Fischer reagent. A moisture content of more than 25 ppm, or nonconformance to 3.8.2 shall constitute failure of this test.

4.5.2.6 Cooling capacity and airflow. Test the air conditioner for cooling capacity at the capacity rating conditions specified in 3.6.2. Test the evaporator fan for airflow at the conditions specified in 3.6.3. Insulating or sealing material that is not inherently a part of the air conditioner shall not be employed.

4.5.2.6.1 Cooling capacity. Capacity tests shall be conducted using the instruments, apparatus, and procedures specified in ASHRAE Standards, and as specified herein. Tests conducted in accordance with Standard 37 shall use air enthalpy method for condenser confirmatory test. Prior to the capacity tests, thermocouples shall be installed at the thermal expansion valve bulb, quench valve bulb, quench valve discharge line, liquid line at the expansion valve inlet, and compressor discharge line. After installation, the thermocouples shall be completely insulated to prevent ambient temperatures from affecting the temperatures obtained from them. In addition to the data required by the applicable ASHRAE standards, temperatures obtained from the above thermocouples shall be recorded at the same time that the ASHRAE data are recorded. These temperatures shall be used, in the event of capacity test failures, to troubleshoot the refrigeration system.

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4.5.2.6.2 Air flow. Air flow tests shall be conducted using the instruments, apparatus, and procedures specified in ASHRAE Standard 37 or ASHRAE Standard 51 (nozzles only for air flow measurement). If desired, AMCA Standard 210 may be used for the airflow tests.

4.5.2.6.3 Basis of failure. Nonconformance to 3.6.2, 3.6.3 or 3.6.6 shall constitute failure of this test.

4.5.2.7 Heating capacity. Position the temperature selection knob at the 12 o'clock position and the mode selector switch on "AUTO". Immerse the conditioned air thermostat bulb in water or air having a temperature of 65 +5 -0 °F. Prior to recording any data, operate the air conditioner for a period of not less than 2 hours with all heater elements energized. Measure the power input (W) to the heating elements and calculate the heating capacity using the following equation:

$$QH = 3.414 \times W$$

where QH = Heating capacity (Btu/hr)
W = Power input (Watts)

Nonconformance to 3.6.4 and 3.6.6 shall constitute failure of this test.

4.5.2.8 High-temperature operation cooling. Subject the air conditioner to a temperature of 125 +3 -0 °F for a period of 4 hours. After completion of the exposure period, operate the air conditioner in the cooling mode for a period of not less than 1 hour in an ambient temperature of 125 +3 -0 °F with air between 125 and 128 °F entering the evaporator and condenser. Inability of the air conditioner to start and operate or nonconformance to 3.6.1.1 or 3.6.7.2 shall constitute failure of this test.

4.5.2.9 High temperature storage. Subject the entire air conditioner to a temperature of 160 °F for a period of 4 hours. At the conclusion of the exposure period, remove the air conditioner from the test chamber and examine for deterioration, corrosion, or damage. After completion of the examination, allow the air conditioner to soak at ambient temperature between 65 and 85 °F for a period of not less than 1 hour but not more than 3 hours. After completion of the soaking period, operate the air conditioner in the cooling mode, at the coolest temperature setting, for a period of not less than 1 hour. Inability of the air conditioner to start, operate, or any evidence of deterioration, corrosion, damage, such as rupture of tubing, damage to control components, binding of motors, fans or other mechanical parts, deterioration of gaskets, damage to insulation, breakdown of adhesives, permanent setting of gaskets or packing, or loss of refrigerant shall constitute failure of this test.

4.5.2.10 Low ambient cooling test. The air conditioner shall be operated with 100 percent return air for 10 hours as specified in table I, condition 4. Any evidence of ice or frost on the

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finned area of the evaporator coil, or failure of the air conditioner to provide cooling as specified in 3.6.2.1 shall constitute failure of this test.

4.5.2.11 Low temperature storage. Subject the air conditioners to a temperature of -60 °F for a continuous period of 12 hours. At the conclusion of the exposure period, remove the air conditioners from the test chamber and examine for deterioration and damage. After completion of the examination, allow the air conditioners to soak at ambient temperature at 65 and 85 °F for a period of not less than 1 hour. After completion of the soaking period, operate the air conditioners in cooling mode at the coolest temperature setting, for a period of not less than 1 hour. Any evidence of deterioration or damage or failure of the air conditioner to start and operate shall constitute failure of this test.

4.5.2.12 Rain. Subject the air conditioner to test method 506, procedure 1 of MIL-STD-810 as specified herein. Rainfall shall be produced by means of water spray and shall be measured with U.S. Weather Bureau type gauges. Nozzles shall emit droplets having a diameter of not less than 1.5 millimeters. Water supply temperature shall be between 50 and 70 °F and shall contain fluorescein dye for ease in determining the locations and extent of leakage. After each rain test, remove panels, access doors, and component covers from the air conditioner as required to examine for water penetration. After the examination, operate the air conditioner for 15 minutes in the cooling mode and 15 minutes in the heating mode.

4.5.2.12.1 Free rain. Close all panels, doors and the fresh air door, install evaporator discharge and return air ducts and seal the open ends. With the air conditioners not operating, subject each of four sides and top of the air conditioners to a simulated rainfall of 4 ±1-inch per hour for 30 minutes each, for a total test duration of not less than 2 hours. Direction of rainfall against each of the four sides shall be varied throughout the test from vertical to 45 degrees from vertical.

4.5.2.12.2 Blowing rain. Open the condenser inlet and discharge doors and remove the end seals from the discharge and return air ducts. With the air conditioner operating in the cooling mode, at the coolest temperature setting, subject each of the four sides to a simulated rainfall of 1 ±0.50-inch per hour combined with a horizontal wind of 40 ±5 miles per hour for a period of 1 hour. Total test duration shall be not less than 3 hours.

4.5.2.12.3 Basis of failure. One or more of the following shall constitute failure of the rain test:

a. Evidence of water leaking into the evaporator or control panel enclosure.

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b. Malfunctioning of components or an electrical short during operation.
c. Nonconformance to 3.6.7.4.

4.5.2.13 Humidity. Subject the air conditioner to test method 507, procedure I, of MIL-STD-810 except the maximum temperature shall be 155 °F. Do not operate the air conditioner during exposure and do not expose the air conditioner to extreme temperature prior to test. Prior to this test and after the exposure period, operate the air conditioner in accordance with 4.5.2.4 to determine proper functioning of the controls. Nonconformance to 3.6.7.3 or 3.9 shall constitute failure of this test.

4.5.2.14 Salt fog. Subject the air conditioner to test method 509, procedure I, of MIL-STD-810. Do not operate the air conditioner during exposure. Prior to this test and after the exposure period, operate the air conditioner in accordance with 4.5.2.4 to determine function of the controls. Nonconformance to 3.6.7.3 shall constitute failure of this test.

4.5.2.15 Vibration. Conduct the vibration test as specified in MIL-STD-810, method 514.4, category 3, procedure III, figure 514.4-4. The test time shall be 60 minutes total per plane. The air conditioners shall not be operated during this test. Auxiliary braces, straps, or other tiedown devices, other than those specified herein, shall not be employed to secure the air conditioners to the test apparatus. No blocking or tiedown of internal components other than those that are part of the air conditioner shall be employed. Removable panels and access doors shall be in place during all testing. With the air conditioner mounted, there shall be no resonance inherent in the test apparatus at the frequencies and acceleration levels specified for this test. During the tests, the base of the air conditioner shall remain in a horizontal plane. Mount accelerometers on the test apparatus immediately adjacent to the four corners of the base of the air conditioner. The four accelerometers shall be connected into an electric or electronic network which will average the accelerometers' output signals into a single indicated g level of the magnitude specified for this test. After completion of each test axis, conduct the controls test specified in 4.5.2.4 and the refrigeration system leak test specified in 4.5.2.3.2.

4.5.2.15.1 Basis of failure. Nonconformance to 3.6.7.6, 3.8.3 and 3.9 shall constitute failure of this test.

4.5.2.16 Run-in.

4.5.2.16.1 Run-in performance. Install ducts on the evaporator-return-air openings, evaporator-air-discharge opening, and condenser air inlet opening of each air conditioner. Each duct shall be not less than 12 inches long and of the same cross-sectional dimensions as the respective openings. Equip

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each duct with a thermocouple grid to measure average entering and leaving air temperatures to the evaporator and entering air to the condenser. Measure the entering evaporator-air wet-bulb temperature by psychrometer device located in the test chamber. Operate the air conditioners for not less than 1 hour in the cooling mode while maintaining the supply voltage to the air conditioner within ±5 percent. The air temperature entering the evaporator return and condenser inlet shall be as specified in table V. After stabilization, three sets of the data specified in table VI shall be recorded and averaged during the last 30 minutes of operation (every 10 minutes). After completion of the above, operate the air conditioner in an ambient temperature between 100 and 120 °F for a total period of 48 hours in the cooling mode, at the coolest temperature setting in cyclic operation of 55 minutes "on" and 5 minutes "off". After completion of the 48-hour operation, continue operating the air conditioner in the cooling mode for not less than an additional hour while maintaining the supply voltage to the air conditioner within ±5 percent and temperature of the air entering the evaporator return and condenser inlet as specified in table V. After stabilization, three sets of the data specified in table VI shall be recorded and averaged during the last 30 minutes of operation (every 10 minutes). With the air conditioner at room ambient temperature, operate the air conditioner on high heat mode for 30 minutes. During the last 10 minutes of operation, record the power input (W) to the air conditioner. After completion of the 30-minute operating period, conduct the refrigerant system leak test as specified in 4.5.2.3.2.

TABLE V. Temperature conditions for run-in performance.

Location	Temperature			
	Condition I	Condition II	Condition III	Condition IV
Evaporator return	87 ±0.5 °F	93 ±0.5 °F	87 ±0.5 °F	93 ±0.5 °F
Condenser inlet	107 ±0.5 °F	113 ±0.5 °F	113 ±0.5 °F	107 ±0.5 °F
Evaporator return and condenser inlet: Dry coil condition, all conditions.				

4.5.2.16.2 Run-in performance curves. Not less than 25 units shall be used to establish acceptance curves. When less than 25 units are produced, all units will be used to establish acceptance curves. Data obtained in accordance with 4.5.2.17.1 shall be used to select one high and one low unit that then shall be capacity-tested at condition 2 of table I. If any high- or low-limit air conditioner fails to meet minimum requirements of 3.6.2, it shall be replaced by the next low or high unit and that unit tested. Operate the acceptable high and low air conditioner continuously in the cooling mode for not less than 1 hour under

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each of the conditions specified in table VII. After stabilization, four sets of data specified in table VI shall be recorded and averaged for each of the conditions in table VII during the last 30 minutes (every 10 minutes). Establish high and low limits of acceptance for production units by plotting the following curves from data specified in table VI:

a. Average evaporator-entering dry-bulb temperature versus average temperature differential between evaporator inlet and outlet air temperatures for each of the condenser-entering temperatures recorded.
b. Average evaporator-entering dry-bulb temperature versus power input (Kw) for each of the condenser-entering temperatures.
c. Average evaporator-entering dry-bulb temperature versus suction pressure for each of the condenser-entering temperatures.
d. Average evaporator-entering dry-bulb temperature versus discharge pressure for each of the condenser-entering temperatures.
e. Average evaporator-entering dry-bulb temperature versus current input (amps) for each of the condenser-entering temperatures.

TABLE VI. Data required for run-in performance curves.

Dry-bulb temperature of air entering the evaporator, °F.
Wet-bulb temperature of air entering the evaporator, °F.
Dry-bulb temperature of air leaving the evaporator, °F.
Dry-bulb temperature of air entering the condenser, °F.
Condensing pressure, psig.
Suction pressure, psig.
Current draw of the air conditioner, amps.
Power consumption of the air conditioner, W.

TABLE VII. Temperature conditions for run-in performance curves.

Location	Temperature			
	Condition I	Condition II	Condition III	Condition IV
Evaporator return	87 ±0.5 °F	93 ±0.5 °F	87 ±0.5 °F	93 ±0.5 °F
Condenser inlet	107 ±0.5 °F	113 ±0.5 °F	113 ±0.5 °F	107 ±0.5 °F
Evaporator return and condenser inlet: Dry coil condition, all conditions.				

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4.5.2.16.3 Failure criteria. One or more of the following shall constitute failure of this test.

a. Temperature differential between the average air dry-bulb temperature entering the evaporator return and the average air dry-bulb temperature leaving the evaporator before and after the 48-hour test phase in 4.5.2.16.1 not within the curves developed as specified in 4.5.2.16.2.
b. Average condensing pressure recorded before and after the 48-hour test phase in 4.5.2.16.1 not within the curves developed as specified in 4.5.2.16.2.
c. Average suction pressure recorded before and after the 48-hour test phase in 4.5.2.16.1 not within the curves developed as specified in 4.5.2.16.2.
d. Average power consumption on cooling recorded before and after the 48-hour test phase in 4.5.2.16.1 not within the curves developed as specified in 4.5.2.16.2.
e. Total power consumption (W) on heating (maximum heat) in excess of that specified in 3.6.6.
f. Failure of any component or fracture or breakdown of any material.
g. Evidence of any detectable refrigerant leak.
h. Need for any servicing of the air conditioner during operation.

4.5.2.17 Reliability. Using the upper and lower test MTBFs specified in 3.4, the air conditioners shall be tested as specified herein with accept and reject criteria in accordance with MIL-HDBK-781, test plan XIV. In addition, the occurrence of more than 2 failures of the same type, or attributable to the same component shall constitute failure of this test. Each air conditioner tested must accumulate at least one third of the total test hours.

4.5.2.17.1 Test facility. Install the air conditioners in a test chamber consisting of two rooms. One room shall function as an indoor room and shall maintain the evaporator return-air temperature within ±5 °F of the temperatures specified in figure 2. The second room shall function as an outdoor room and shall maintain the condenser return air temperature within ±5 °F of the temperatures specified in figure 2. These temperatures shall be maintained on a 24-hour cyclic basis throughout the reliability test, as shown on figure 2.

4.5.2.17.2 Operation. The mode selector switch and temperature selection control shall be operated manually or automatically. If automatic means are used, they shall not interrupt the control circuit but shall physically operate the air conditioner control components. The controls shall be positioned as shown in table VIII. The air conditioner shall be operated in accordance with the standardized test profile shown in figure 2 for the duration specified in 4.5.2.17.

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TABLE VIII. Control positions for reliability test.

Mode	Time	Control	Control Position
Heating	0030 to 0130	Mode Switch	Auto
Heating	0030 to 0100	Thermostat *	Warmest
Heating	0100 to 0130	Thermostat *	Coolest
Vent	0130 to 0400	Mode Switch	Vent
Vent	0130 to 0400	Thermostat *	Warmest
Cooling	0400 to 2100	Mode Switch	Auto
Cooling	0400 to 2100	Thermostat *	Coolest
Cooling	2100 to 2400	Mode Switch	Auto
Cooling	2100 to 2400	Thermostat *	Warmest
Off	0000 to 0030	Mode Switch	Off

* Temperature selection control.

4.5.2.17.3 Data. The following shall be monitored and recorded throughout the test as required to determine that normal air conditioner operation is being maintained and to identify problems in the event of failure or malfunction:

a. Voltage.
b. Current.
c. Frequency.
d. Power factor.
e. Refrigerant pressures.
   (1) Suction.
   (2) Discharge.
f. Refrigerant temperatures.
   (1) Compressor discharge.
   (2) Liquid line.
   (3) Suction line (at compressor).
g. Air temperatures.
   (1) Evaporator return (dry bulb).
   (2) Evaporator discharge (dry bulb).
   (3) Condenser return (dry bulb).
h. Abnormal operation (visual and audio observation).
i. Sight-glass observation (clear or bubbles).
j. Identification of unit.
k. Initials of technician.
l. All failures.
m. Time of day of failure.
n. Action taken if failure occurs.

4.5.2.17.4 Failure criteria. A failure is defined as any malfunction for which corrective action cannot be deferred until the next scheduled routine maintenance check, provided corrective maintenance action can be performed at the organizational level with available tools, or which cannot be corrected by the operator crew with available tools within 30 minutes. Corrective action is not deferrable if the malfunction causes or may cause inability to commence operation of the unit, cessation of

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operation of the unit, or degradation of performance capability as specified herein. Malfunctions will not be counted as "failures" for reliability purposes if detected and corrected during initial inspection or if they may be deferred to final inspection (except for those that cause test termination). Malfunctions that result from operational or maintenance errors will not be counted. A malfunction shall be considered to have occurred if the following happens:

a. Temperature differences across the evaporator coil is less than ±5 °F (dry bulb) than the values determined in table VI when the air conditioner is in the cooling mode.
b. Current draw is greater than ±2 amperes than the values determined in table VI when the air conditioner is in the cooling mode with the temperature selection control in the coolest position.
c. Current draw does not decrease when the air conditioner is in the heating mode and the temperature selection control is changed from the warmest position to the coolest position.
d. Temperature at the evaporator discharge does not increase when the air conditioner is turned to the heating mode with the temperature selection control set in the warmest position.
e. The sight-glass is not clear when the air conditioner is in the cooling mode.
f. The suction pressure is lower than ±5 psig than the values determined in 4.5.2.16.2.c when the air conditioner is in the cooling mode.
g. The discharge pressure is greater than the values determined in 4.5.2.16.2.d when the air conditioner is in the cooling mode.
h. Any abnormal operation occurs as determined by looking at or listening to the unit.

4.5.2.18 Maintenance evaluation. The maintenance ratio shall be computed as specified in 3.4.2 for all maintenance actions required during or as a result of first article testing. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2.19 Safety. Evaluation shall be accomplished throughout testing to determine conformance to MIL-STD-882. Nonconformance to 3.16 shall constitute failure of this test.

4.5.2.20 Human factors engineering. The air conditioner shall be evaluated to determine that the human factors criteria have been met. Nonconformance to 3.12 shall constitute failure of this test.

4.5.2.21 Noise level test. The acoustic noise levels produced by the air conditioner shall not exceed the sound pressure level values specified in 3.12.1 at any of the octave bands or measurement positions specified in 4.5.2.21.2. All acoustic

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measurements shall be made with the air conditioner operating at maximum rated capacity. The acoustic measurement instrumentation shall be as specified in 4.5.2.21.3.

4.5.2.21.1 Background noise. Acoustic measurements shall be made only when the sound pressure levels produced by the air conditioner are more than 10 dB above the ambient background sound pressure level in such octave band.

4.5.2.21.2 Measurement positions. Acoustic measurements shall be made at 12 positions equally spaced every 30 degrees on 10-foot radius circle centered on the geometrical center of the air conditioner. All measurements shall be made with the air conditioner located on level unobstructed pavement with the microphone 5 feet above the pavement level and oriented toward the air conditioner. A flexible duct shall be attached to the outlet of the air conditioner and laid out to discharge the air away from the air conditioner.

4.5.2.21.3 Instrumentation and measurement. Noise measurement shall be conducted in accordance with MIL-STD-1474, 5.1.2.5.2 and 5.1.2.5.3.

4.6 Inspection of packaging.

4.6.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. One or more defects shall be cause for rejection.

130.	Any removable components or accessories not placed in their designated storage areas of the air conditioners and doors not closed (see 5.2.1).
131.	Technical publications not preserved and secured to the air conditioners as specified for level A (see 5.2.1.1).
132.	Air conditioners not preserved as specified for level A (see 5.2.1.1.2).
133.	Commercial preservation not as specified by the reference document (see 5.2.1.2).
134.	Shipping containers not as specified for levels A and B (see 5.3.1 and 5.3.2).

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135. Strapping not as specified for level A or B (see 5.3.1 and 5.3.2).
136. Blocking and bracing not as specified for levels A and B (see 5.3.1 and 5.3.2).
137. Packing for commercial not as specified by the referenced document (see 5.3.3).
138. Marking not as specified for military or commercial (see 5.4.1 and 5.4.2).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The first article pack may be accomplished utilizing either the first article model air conditioner or a production air conditioner. If the first article model air conditioner is utilized, any preservation, and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model air conditioner and production air conditioners.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 General. Any removable components or accessories such as duct adapters, power cables, drain hoses, etc., shall be placed and secured in their designated storage areas within the air conditioners. All doors shall be placed in their closed position, when applicable.

5.2.1.1 Level A.

5.2.1.1.1 Technical publications. Technical publication shall be preserved by MIL-P-116, method IC-1 or IC-3 and secured to the air conditioners within the barrier material of method II specified in 5.2.1.2 and prepared for shipment as specified in 5.2.1.

5.2.1.1.2 Air conditioners. Each air conditioner shall be preserved in accordance with MIL-P-116, method IIA.

5.2.1.2 Commercial. Each air conditioner, prepared for shipment as specified in 5.2.1, shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

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5.3.1 Level A. The air conditioner preserved as specified in 5.2 shall be packed in overseas-type containers conforming to PPP-B-601. The air conditioner shall be cushioned, anchored, blocked and braced in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the appendix of the box specification and strapping shall be zinc-coated.

5.3.2 Level B. Level B packing shall be the same as level A, except the containers shall be of the domestic type and strapping may be zinc-coated.

5.3.3 Commercial. The air conditioner preserved as specified in 5.2 shall be packed in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Military. In addition to any special marking required by the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951.

6. NOTES
(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The air conditioner is intended for use in cooling, heating, dehumidifying, filtering, and circulating air for portable shelters or vans in accordance with electronic and personnel requirements. The air conditioner is designed to be used while being directly exposed to the atmosphere.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title number and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required for inspection and approval; and when required, the number of units required, and time frame for submittal (see 3.2).
d. Location and conditions for first article testing (see 3.2 and 4.3).
e. In the event it becomes necessary to change said drawings in order to achieve the performance requirements of this specification the contractor shall immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes will be processed in accordance with the provisions of the changes clause of the contract (see 3.3).

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f. Color required when other than camouflage green (see 3.15.1).
g. Time frame required for submission of first article pack (see 5.1).
h. Degree of preservation and packing (see 5.2 and 5.3).
i. Any special shipment marking (see 5.4).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of five units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 Preliminary run-in log. The contracting officer should arrange for the contractor to maintain a certified log of all preliminary running times as specified in 4.5.2.16.

6.3.2 Readings. Readings of all pertinent pressures, temperatures, rotational speeds, and other data necessary in demonstrating compliance with the applicable portions of section 3 shall be recorded at intervals not exceeding 30 minutes during all tests specified herein.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.3	DI-T-4901	First article inspection procedure	(see 6.5)
4.3	DI-T-4902	First article inspection report	(see 6.5)

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The DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.5 Reliability and maintainability information. The following information shall be included as an attachment accompanying the first article test report (same quality and quantity of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, cycles, or position in the test procedure as appropriate. Also, test conditions during failures or irregular operating shall be recorded.
b. Description of the engineering reasoning behind any tests conducted to determine assignable causes for all failures and irregular operations.
c. Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and reasoning behind the predicted effectiveness of those corrections.
d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.
e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of the obligation to preserve, pack, and mark the air conditioners in accordance with the specification.

6.7 Definitions. For the purpose of this specification, the following definitions will apply:

6.7.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

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Confidence level: 90 percent

$$MTBF = \frac{Total\ test\ time}{2.3}$$

$$Reliability - e^{-x}\ where\ x - \frac{Mission\ Time}{MTBF}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

6.7.2 Mean-time-between-failures (MTBF). The MTBF is the arithmetical mean (average) of the operating time between failures.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Subject term (key word listing).

Air filtration device
Heater, air
Unit, control, environmental

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4120-A330

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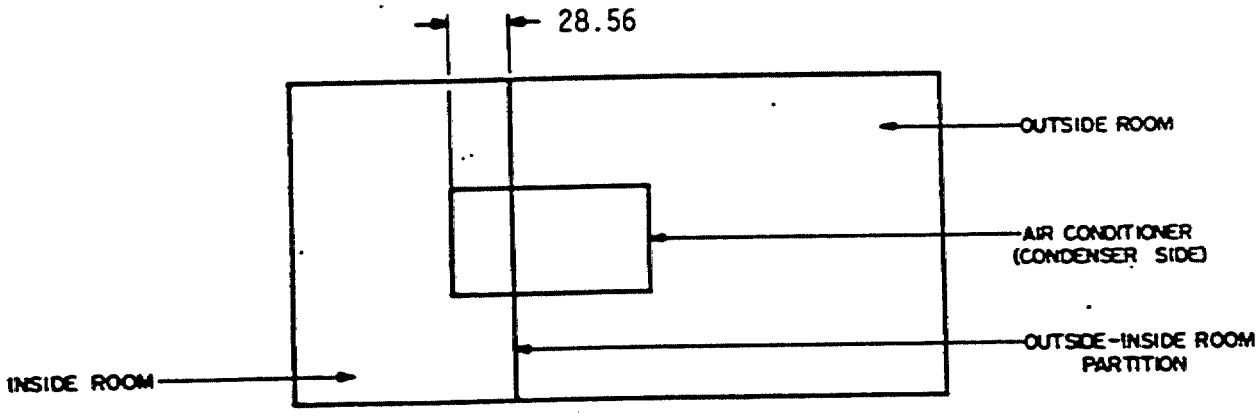


FIGURE 1. Air conditioner location for capacity test.

X-2440A

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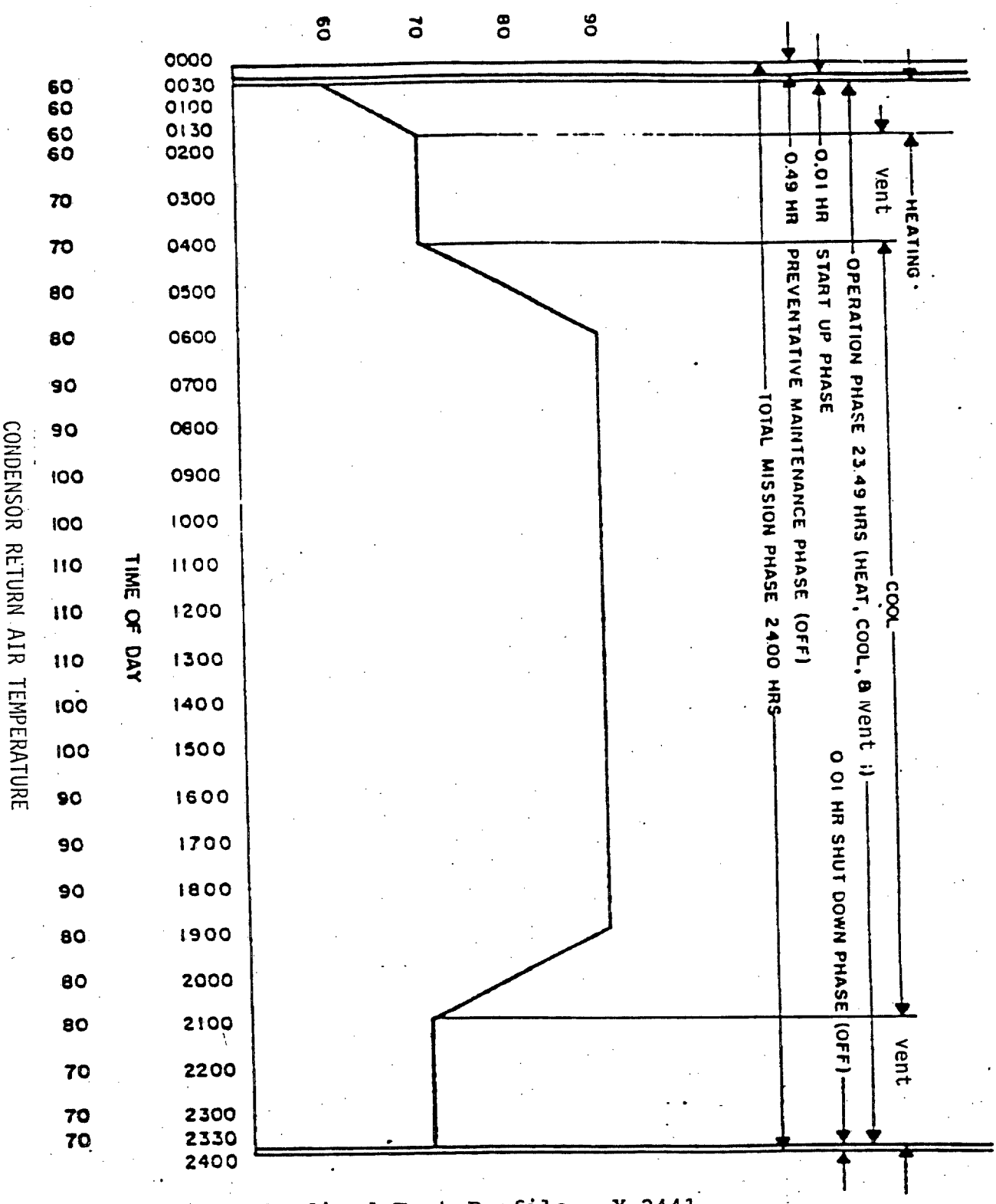


FIGURE 2. Standardized Test Profile. X-2441

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-0083216C(ME)	2. DOCUMENT DATE (YYMMDD) 940805

3. DOCUMENT TITLE
AIR CONDITIONER, HORIZONAL, 54,000 BTUH, 208/230V

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) DSN 654-3468
c. ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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